Subject: ASD&D	**Semester:** V
Class: TE	**Batch:** T11
Name: Irisha Deshpande	**Roll No:** 2401022

Experiment 6

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

Lab Outcome: Mapped LO1
LO1: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
    - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., *Core Infrastructure*).
    - Story: A end-user feature written from a user perspective (e.g., *Storefront UI*).
    - Task / Bug: Technical implementations or defects to resolve.
    - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (*To Do*, *In Progress*, *In Review*, *Done*).

- Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

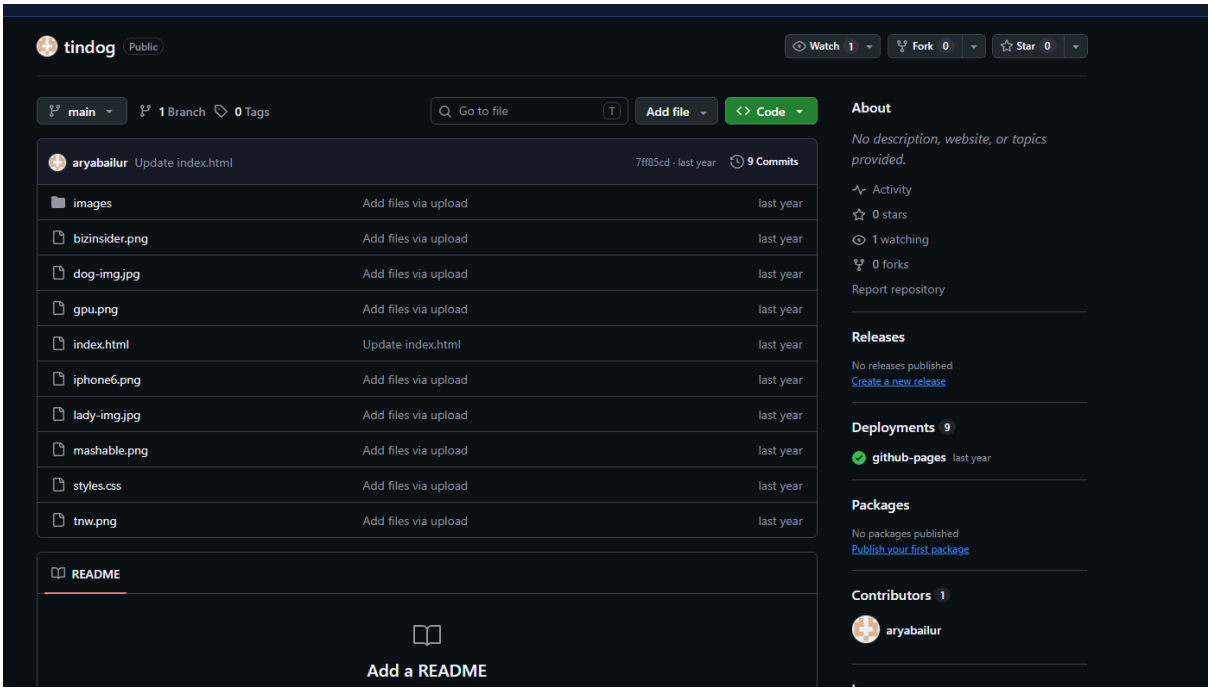
- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

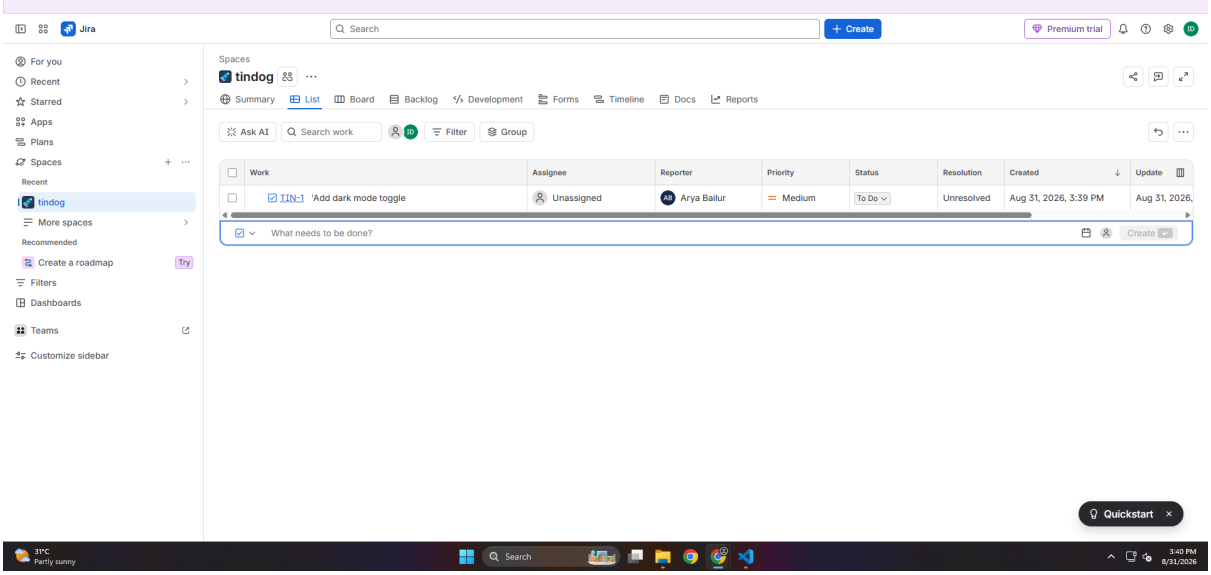
DevOps Integration & Automation Lifecycle

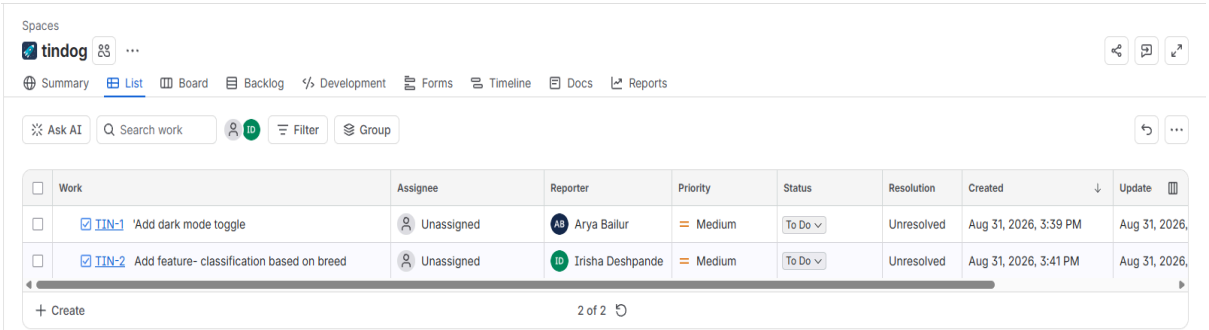
- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or Bitbucket) through official marketplace apps (e.g., *GitHub for Jira*).
- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.
- Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to *In Review* when a Pull Request is opened, or transitioning to *Done* upon a merged PR).

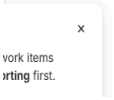
Team Collaboration & Access Management

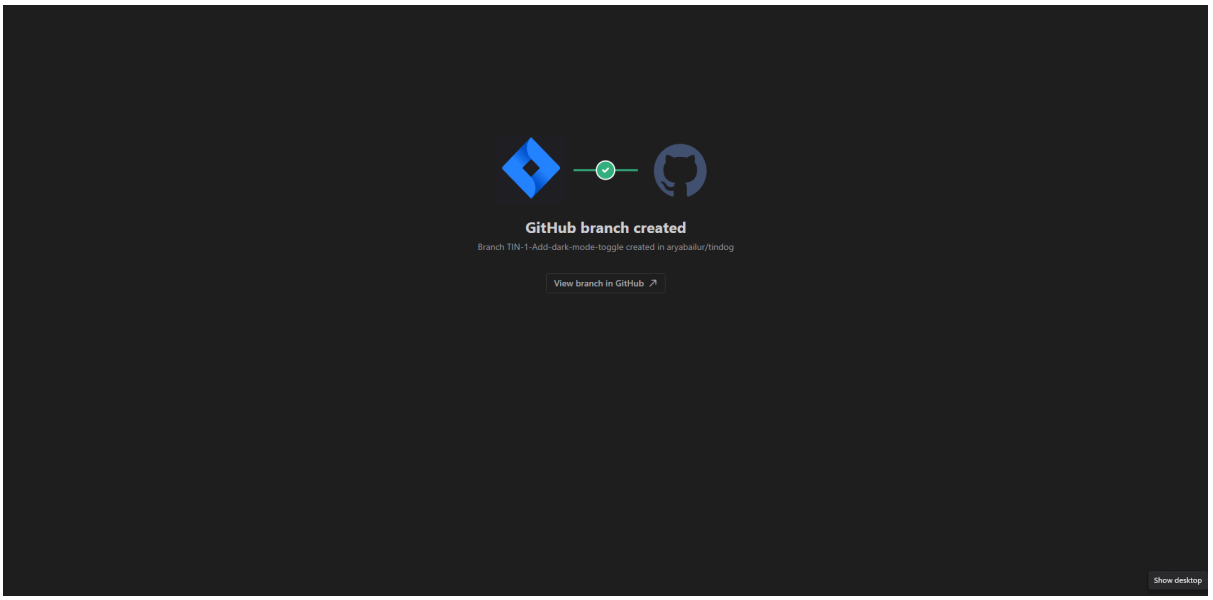
- Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (*Administrator*, *Member*, *Viewer*).
- Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.

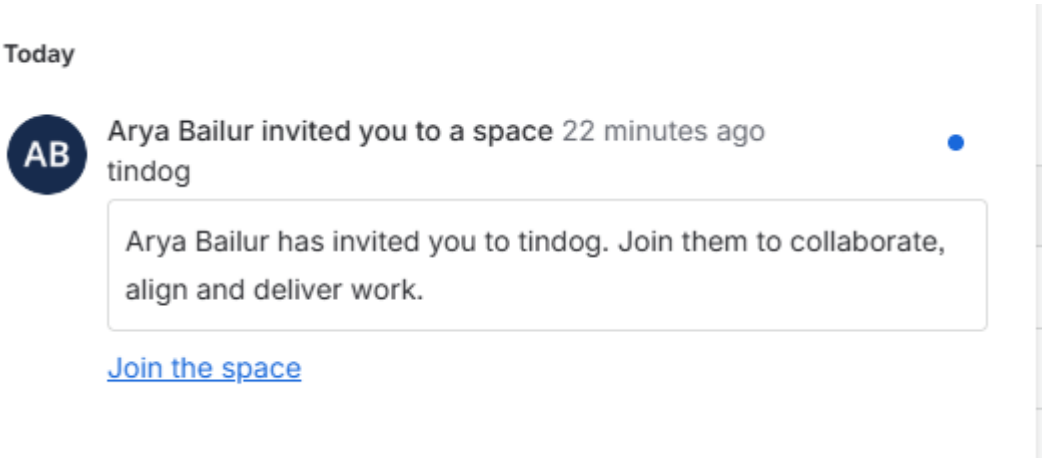

tindog
Public
Watch 1
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
aryabailur Update index.html
7ff85cd · last year
9 Commits
images Add files via upload last year
bizinsider.png Add files via upload last year
dog-img.jpg Add files via upload last year
gpu.png Add files via upload last year
index.html Update index.html last year
iphone6.png Add files via upload last year
lady-img.jpg Add files via upload last year
mashable.png Add files via upload last year
styles.css Add files via upload last year
tnw.png Add files via upload last year
README
Add a README
About
No description, website, or topics provided.
Activity
0 stars
1 watching
0 forks
Report repository
Releases
No releases published
Create a new release
Deployments 9
github-pages last year
Packages
No packages published
Publish your first package
Contributors 1
aryabailur


Jira
Search
Create
Premium trial
For you
Recent
Starred
Apps
Plans
Spaces
Recent
tindog
More spaces
Recommended
Create a roadmap
Try
Filters
Dashboards
Teams
Customize sidebar
Spaces
tindog
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Ask AI
Search work
Filter
Group
Work
Assignee
Reporter
Priority
Status
Resolution
Created
Update
TIN-1 'Add dark mode toggle
Unassigned
Arya Bailur
Medium
To Do
Unresolved
Aug 31, 2026, 3:39 PM
Aug 31, 2026,
What needs to be done?
Create
Quickstart
31°C
Partly sunny
Search
3:40 PM
8/31/2026


Spaces
tindog
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Ask AI
Search work
Filter
Group
Work
Assignee
Reporter
Priority
Status
Resolution
Created
Update
TIN-1 'Add dark mode toggle
Unassigned
Arya Bailur
Medium
To Do
Unresolved
Aug 31, 2026, 3:39 PM
Aug 31, 2026,
TIN-2 Add feature- classification based on breed
Unassigned
Irisha Deshpande
Medium
To Do
Unresolved
Aug 31, 2026, 3:41 PM
Aug 31, 2026,
Create
2 of 2


vork items
orting first.


GitHub branch created
Branch TIN-1-Add-dark-mode-toggle created in aryabailur/tindog
View branch in GitHub
Show desktop


Today
AB
Arya Bailur invited you to a space 22 minutes ago
tindog
Arya Bailur has invited you to tindog. Join them to collaborate, align and deliver work.
Join the space

Conclusion:

Integrating Jira with GitHub established an automated, end-to-end Agile development lifecycle for our E-commerce application. We learned how to organize backlogs into active sprints, assign tasks across team members, and track real-time progress using Scrum boards. By referencing issue keys in branch names and Pull Requests, we enabled seamless traceability between our codebase and Jira tickets. Furthermore, setting up GitHub Actions CI pipelines and Jira Automation demonstrated how modern DevOps workflows eliminate manual overhead by automating testing and board transitions upon code merges.